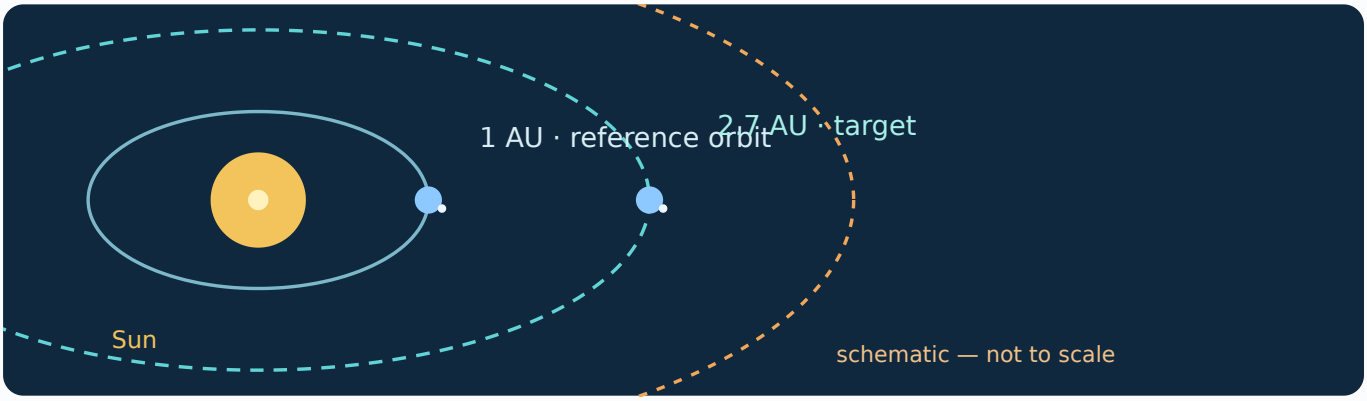
A PHYSICS-LED PLANETARY-MOBILITY PROTOTYPE

Move the orbit. Keep the accounting.

OCTO-EARTH studies a hypothetical move of the Earth–Moon system from 1 astronomical unit toward a 2.7 AU circular orbit, with a simplified climate-control objective. Earth moves only through reaction-mass ejections; momentum and energy bookkeeping are explicit.

What this is—and is not

A runnable orbital/RL simulation and visual explanation. It is not a real propulsion design, a certified structure, a climate forecast, a current PPO success report, or evidence that planetary relocation is feasible.



≤10,000
simulated years · challenge planning horizon

288 K
internal climate target in the hypothetical preset

3
individual enablers · support systems separated

Current evidence: 46 software tests, 22 physics checks and 10 supplied-guide benchmarks passed on 27 Oct 2026. No current-version PPO retraining/evaluation or hardware certification has been performed.

Three selected enablers—three single assumptions

Each selected enabler names one mission-critical resource or technology. Supporting equipment, feedstock and assumptions are listed separately; this is a classification boundary, not a claim that the hardware is ordinary or buildable.

ID / individual enabler	Why selected / model treatment	Boundary and gap
E1 · Access to a finite 3×10^{22} kg asteroid	Provides the modeled reaction-mass reservoir. Packet mass is finite; recoil and mass/energy bookkeeping are tracked.	No capture, docking or placement of this body is simulated. Access is far beyond demonstrated capability.
E2 · Thin-film photovoltaic collector field	Modeled electrical source: 5 million collectors at 0.5 AU intercept the configured 30% of solar output. Conversion and downstream energy ledger are represented.	Area, mass, manufacture, deployment and industrial scale are unverified. Power transfer is separate support equipment.
E3 · Space-mirror ring	Redirects sunlight to the simplified climate model at the target orbit. Mirrors create no energy.	Real orbit, formation flight, pointing, local illumination and structural loads are not validated.

Separate support for E1

- Earth-adjacent launch ring and eight 2-Earth-radius mass-driver arms; one packet per joint throw.
- Arm power conditioning and finite reserve.
- Directional arm radiators and assumed non-damaging operation—neither physically designed.

Separate support for E2

- Microwave phased arrays and arm-tip rectennas.
- Collector sail-tilt station keeping and clear sunlight lanes.
- Factories, Mercury feedstock and construction/repair logistics.

Separate support for E3

Terminator-orbit placement, formation flight, attitude control and pointing. These are not part of the selected mirror-ring enabler label and remain unverified.

What is not hidden: energy originates in modeled sunlight; momentum originates in ejected mass; mirror light is reflected solar energy. No reactionless thrust and no direct RL velocity edit.

Target, tolerances and caveats

The challenge asks for an outward transfer under safety and structural constraints. The project's internal windows help evaluate a simulation; they are not substitutes for the event's complete rulebook.

Quantity	Project value	Correct interpretation
Target semimajor axis	2.7 AU, ±0.05 AU	Project internal target window.
Orbital-plane inclination	≤5°	Angle of the orbital plane relative to the reference plane. It is not Earth's spin-axis tilt and not 5%.
Final eccentricity	$\|e\| \le 0.05$	Absolute final eccentricity tolerance; not a cap on the amount e changes.
Climate set point	288 K ±5 K	Global-mean two-box model setting, not a full habitability or life criterion.
Active hold	200 simulated years by default	Agent remains in control; a breach resets the project hold clock.
Passive coast	Separate high-fidelity check	Zero-ejection coast after evaluation; not active control and not PPO feedback.

Rotation-axis tilt: the reviewed PDFs do not specify a numeric rotation-axis-tilt limit or eccentricity-change limit. The model does not simulate axial tilt, spin, seasons or tides/heating. No official threshold is invented here.

Key kinematic and energy estimates · hypothetical preset

11,658 m/s
Slow-spiral Δv estimate, 1 → 2.7 AU.

```
ΔE orbit ≈ 1.697×10³³ J
```

1.664×10^{35} J
Modeled minimum launcher energy draw.

```
Peak launcher demand ≈ 6.338×10²⁴ W
```

1.415×10^{22} kg
Reaction mass required at the configured maximum launch speed, from a finite 3×10^{22} kg reservoir.

```
≈0.193 lunar masses required
```

1.114×10^{25} W
Estimated minimum electrical delivery at 2.7 AU.

```
Calculated configuration, not hardware
availability
```

Feasibility label: the analytic gate says "feasible within this model" for this extreme hypothetical preset. That label is not real-world feasibility, mission acceptance, or an engineering proof.

Where the numbers come from

The ledger keeps energy sources and sinks explicit. The PV field powers the separate microwave chain; the mirror ring reflects sunlight and is not a power generator.

Energy path

Sunlight → photovoltaic conversion → microwave transmitters → diffraction/capture → arm-tip rectennas → launcher electricity → packet kinetic energy. The books also show collector heat, manufacturing, transmitter heat, beam spill, rectenna heat, launcher heat and any reserve used.

Subsystem	Hypothetical configuration estimate	Important limit
PV collector field (E2)	5,000,000 panels; total area ≈2.109×10^{22} m²; square side ≈64,950 km per panel; total panel mass ≈2.109×10^{21} kg (≈2.87% lunar mass).	0.5 AU; configured interception ≈30% of solar output. Huge field, not deployable hardware.
Electrical delivery	≈1.290×10^{25} W at 1 AU; ≈1.114×10^{25} W at 2.7 AU.	Includes modeled efficiency, temperature, aging, beam capture and rectenna efficiency. Transfer components are separate support systems.
Space-mirror ring (E3)	Area 1.6×10^{15} m²; reflective capacity ≈1.901×10^{17} W at 2.7 AU (≈1.10× present Earth-intercepted sunlight).	90% reflectivity and 5 g/m² configured film; pointing and local illumination are simplified.
Energy availability screen	Estimated 10,000-year electrical availability ≈3.817×10^{36} J.	Not evidence of constructed collectors, perfect uptime or practical beaming.

Accounting checks

- Energy generated by panels is distinct from Earth–Moon orbital energy and from launcher draw; do not conflate these boundaries.
- Each ejection has a finite packet, launch-energy budget and equal-and-opposite recoil. Fragments are tracked in the high-fidelity model.
- The mirror ledger separates sunlight incident, light delivered to Earth, light directed past Earth and mirror heat.
- Reward is not a physical metric. The report presents orbit, energy, mass and climate outputs separately.

Scale warning: 1.114×10^{25} W is an estimated electrical delivery at the target, while 6.338×10^{24} W is peak launcher demand. Neither is a measured source nor proof that power electronics or beams can operate safely.

Control is downstream of physics

Fast model · training

Hierarchical Kepler propagation advances the Earth–Moon barycentre and Moon orbit; ejections exchange momentum with a finite reservoir. Moon gravity is included, while the Sun's tidal effect on the Moon is approximated/omitted in this model. Planetary perturbations use documented approximations.

Used for PPO experience collection; not a replacement for high-fidelity checks.

High-fidelity model · physical check

N-body Sun/Earth/Moon with explicit fragments, event handling and energy, momentum, angular-momentum and mass ledgers. Mercury, Venus, Mars and Jupiter are included; planets beyond Jupiter are omitted.

"High-fidelity" is relative to the fast model, not a claim of real-world mission fidelity.

Policy loop

31 observations → PPO policy → 5 actions (throttle, direction angles, ejection speed, mirror-light set point) → physical transition → new orbit, mass, energy, safety and climate metrics. Reward is logged separately from those physical metrics. The agent cannot directly edit Earth velocity or create free energy.

Arrival is not the finish

1 · ACTIVE HOLD

Agent continues making decisions. Orbit and enabled climate must stay in tolerance for the configured consecutive simulated-year hold; a breach resets progress.

2 · PASSIVE COAST

Separate high-fidelity zero-ejection coast after evaluation. The agent does not control this phase; it does not feed back into PPO training.

RL evidence status: current environment is `controlled-target-hold-v1`. No PPO retraining or current-version target-hold evaluation has been run. Legacy checkpoint and first-capture metrics are historical only; no current RL success is claimed.

Core code responsibilities

`octoearth_rl.py` contains configuration, feasibility checks, fast/high-fidelity engines, Gymnasium environment, observation/action/reward mapping, baselines, training/evaluation hooks, conservation checks, guide benchmarks, reports and the API serving the viewer. The PPO trainer is not the physics engine.

Separate simulated state from explanatory art

View / output	Driven by simulation state	Illustrative or simplified layer
Solar-system overview	Recorded body states, orbital elements, ejection events and telemetry; time bar/speed control use the model history.	Overview scale is compressed; Moon/body sizes may be enlarged for visibility. Distances are labeled schematic where compressed.
Launcher / fragments	Packet velocities, recoil, fragment fates and mass/energy ledger are recorded by physics.	Arms and structural behavior are geometry/visual explanations; no real segment stress or launch shock simulation.
Beams / energy panel	Displayed power follows modeled photovoltaic, transfer, capture and ledger outputs.	Beam visualization is not a physical transmission or receiver demonstration.
Collector close-up	Tracks a small representative set of collectors in the station-keeping study.	Only 12 collector trajectories are followed; not all five million hardware units or their construction.
Earth POV / Weather & Life	Climate temperature, ice-line and growing-area indicators come from the two-box model; Earth-POV layers are individually selectable.	Clouds, winds, storms, aurora/environment layers are visual context; they do not feed back into orbit or climate.

Three failed approaches, not bundled into the main launcher

FAILED 1

Laser + Dyson collector

Required beam ≈6.614×10^{25} W; simple surface estimate ≈43,993 K. Merged laser/radiation-pressure concept.

FAILED 2

Hydrogen harvest

Ideal fusion screen needs ≈1.949×10^{21} kg; even all solar wind for 10,000 y supplies ≈24%.

FAILED 3

Lorentz arms

Required ≈4.351×10^{17} A/arm vs ≈2.25×10^{8} A quench screen; internal forces cancel.

Read the views honestly: animation is not validation. Use labels such as “simulated”, “model estimate”, “schematic” and “illustrative” to keep the evidence boundary visible.

What passed—and what that proves

46/46
automated tests passed

22/22
physics self-checks passed

10/10
participant-guide benchmarks passed

Guide benchmarks cover: Lorentz force; closed-loop force in uniform field; Biot–Savart field; eccentric Kepler orbit; five-year energy baseline; Earth–Moon barycentre period; Hohmann transfer; tangential orbital push; cantilever formula screen; blackbody equilibrium.

Structural floor: the guide requires safety factor ≥1.5 for every structural element. Formula-level screens pass their test cases, but the main eight-arm launcher has no material, cross-section, hinge, buckling, dynamic or fatigue analysis and no structural pass/certification.

Planetary safety: Mars gravity and orbit are represented; the configurable shield uses internal tide, Moon-orbit-perturbation and minimum-distance screens (default safe distance ≈0.0246 AU). These are adjustable project settings, not official safety standards; other rulebook thresholds were absent from the supplied PDFs.

Guide judging weights

Criterion	Marks	Evidence to show
Physical correctness	30	Reaction, energy/momentum, mass and loads.
Validation / numerical rigour	20	Benchmarks, convergence, baselines and tolerances.
Scientific honesty	15	Enablers, assumptions, failures and gaps.
Simulation depth	15	Orbits, depletion, fragments, habitability proxy.
Visualization / usability	10	Readable labels, selectable views, telemetry.
Presentation / questions	10	Estimate, forces, ledger, safety and next tests.
Optional energy bonus	+10	Only if a real modeled source meets the need in time; no automatic credit.

Challenge format: demo video ≤3 minutes · pitch 7 minutes · Q&A 5 minutes · technical report ≤8 pages. No energy bonus or hardware compliance is claimed by this prototype.

Open gates before any feasibility claim

Unresolved engineering

- Asteroid capture, docking, placement and planetary-scale mass handling.
- Launcher structure, segment/hinge loads, packet acceleration, shock, atmosphere and Earth surface effects.
- Directional radiators, extreme waste heat, rectenna intensity and beam spill.
- Photovoltaic deployment, industrial construction, Mercury mining, transmission and maintenance.
- Mirror formation/orbits, pointing errors, illumination pattern and life-cycle material supply.

Unresolved model validation

- Current PPO retraining, multi-seed target-hold evaluation and passive-coast evaluation.
- Long-term orbit stability near Jupiter resonances and omitted outer-planet effects.
- Regional climate, atmosphere, oceans, water cycle, biology, spin and seasons.
- Full structural SF ≥1.5 check and additional rulebook safety limits absent from the supplied PDFs.
- Instrumentation, faults, communications, control delay and operational resilience.

Next evidence gate

Retrain only after confirming the current environment; evaluate held-out seeds through continuous active hold and the separate passive high-fidelity coast; publish orbital, safety, mass, energy and climate metrics separately from reward; then connect real structural, thermal and launch-shock analyses.

Presentation and source references

Supplied: *Planetary Propulsion Hackathon Participant Guide* (pp. 2–12, including benchmark/judging sections) and *TF'26 Problem Statements* (Octo-Earth pp. 42–43). The crosswalk is in `frontend/physics_guide.html`; the executable source is `octoearth_rl.py`; failed concept calculations are in `failed_models.py`; full calculations are in `CALCULATIONS_REPORT.pdf/.html`.

The reviewed PDFs do not include numeric rotation-axis-tilt or eccentricity-change thresholds. Project-internal values (inclination ≤5°, absolute final e≤0.05, 2.7 AU ±0.05 AU and a 200 simulated-year hold) must not be presented as official rules. Every headline number in this brief is an estimate from a hypothetical model preset.

> **Bottom line:** OCTO-EARTH contributes an inspectable physics-and-control prototype, explicit accounting, three separately bounded assumptions and three documented failures. It does not demonstrate a feasible planetary propulsion system.